Data for Social Good

SafeRoads US — Data-Driven Road Safety Insights

Transforming 7.7 million traffic incidents into clear, actionable insights for safer roads and communities.

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Overview

Project Context & Objectives

Data Challenge

7.7M accident records across 49 US States with weather, spatial, and temporal dimensions.

Advanced Statistics

Hypothesis testing, OLS regression diagnostics, and PCA dimensionality reduction

Public Safety Impact

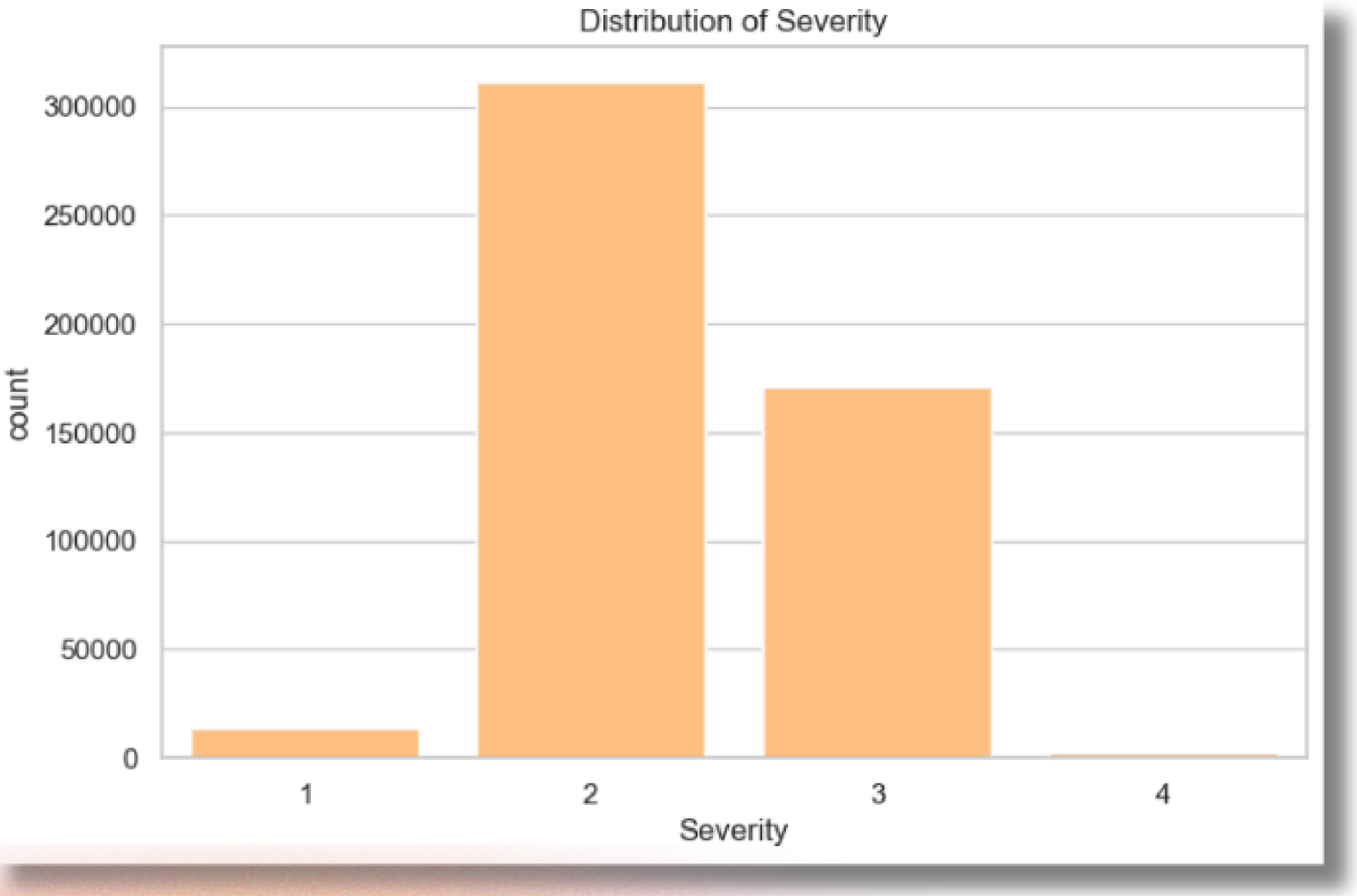
Translating analytical findings into actionable risk awareness for driver safety.

Behind 7.7 million records lies a critical mission: turning raw data into actionable public safety strategies.

Accident Severity Distribution: The Everyday Reality

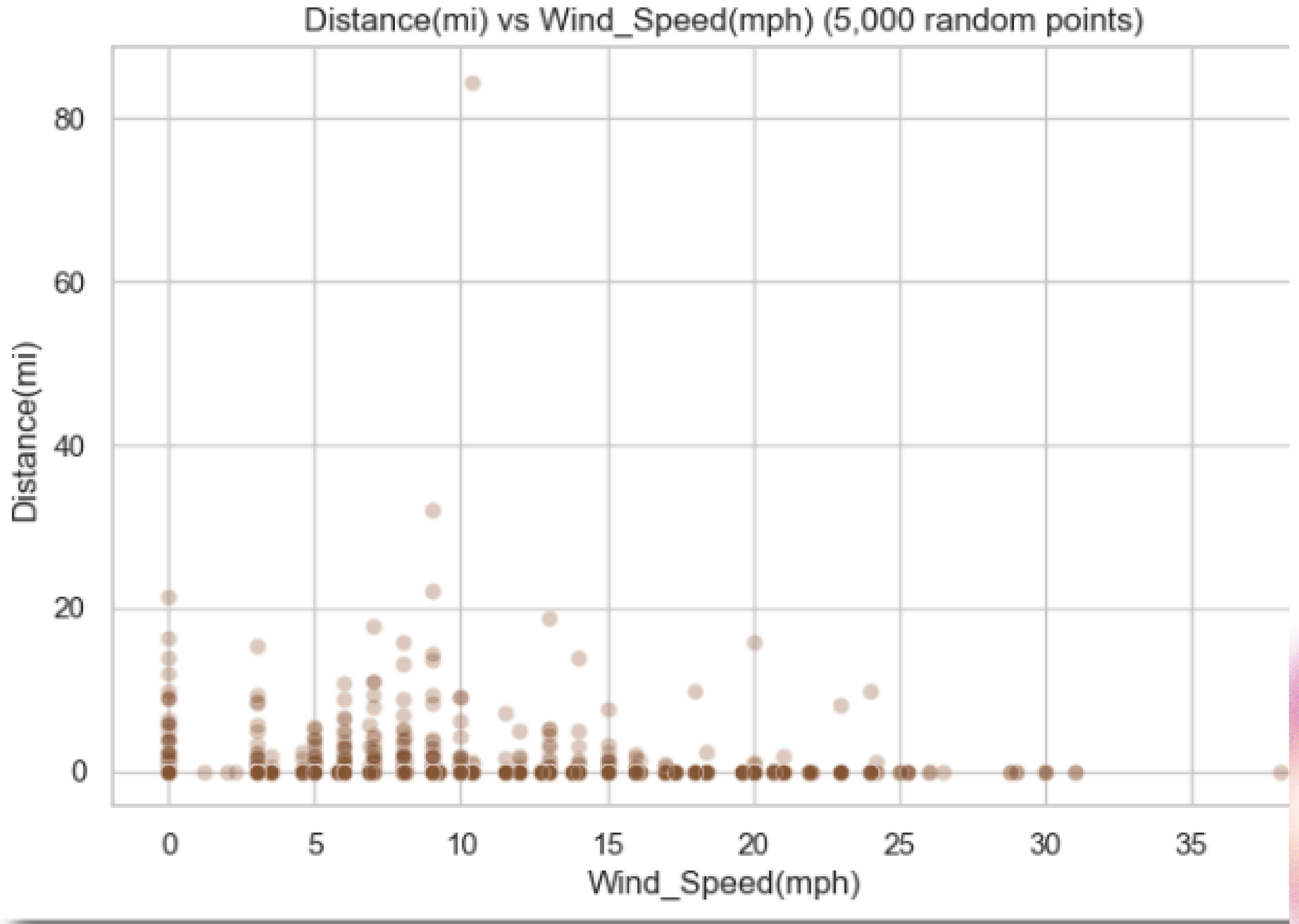
96.7% Central Concentration: Over 96.7% of recorded accidents fall into Severity Levels 2 and 3 (62.43% and 34.27% respectively), showing that moderate traffic friction dominates road safety challenges.

Class Imbalance Challenge: Severe level 4 accidents account for only 0.53% of all cases, presenting a heavy class imbalance that requires sampling strategies in predictive modeling.



Takeaway: "Road safety interventions must focus heavily on high-frequency, moderate-impact events rather than only catastrophic scenarios."

Weather Myths vs. Reality: Wind Speed Analysis

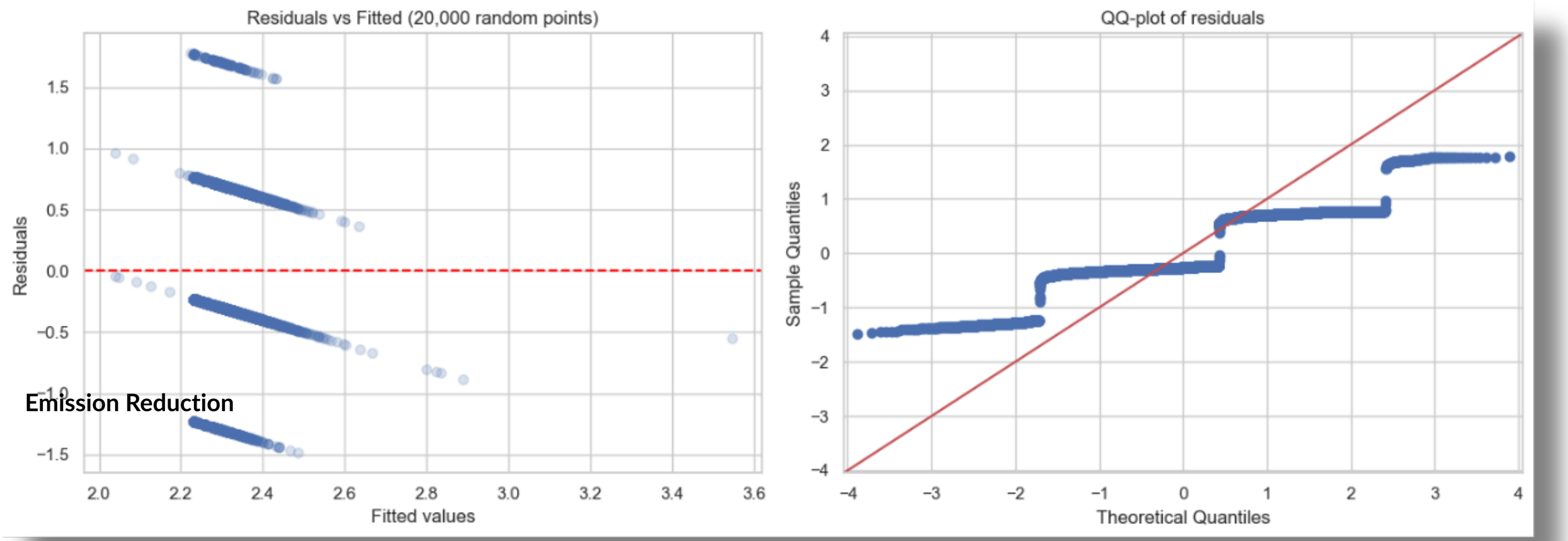


No Direct Link: Statistical testing reveals a Pearson correlation of $r = 0.0019$ ($p = 0.1754$), proving no linear relationship between wind speed and impact distance.

Dispersion Insights: High-impact accidents occur even during low-wind or clear weather conditions.

Takeaway: "Driver behavior and road infrastructure play a far greater role in accident impact distance than isolated weather metrics like wind speed."

Evaluating Weather Variables with Linear Regression



Emission Reduction

Modeling severity against Visibility, Wind Speed, and Precipitation yields $R^2 = 0.008$, confirming weather metrics alone cannot predict severity levels

Residual diagnostics (QQ-plots and clear parallel bands) demonstrate that traditional OLS linear models are structurally limited for discrete severity scores.

Moving beyond linear regressions to non-linear or classification algorithms (e.g., Random Forest, XGBoost) is essential for accurate modeling

The Real-World Impact of Road Incidents

ANNUAL INCIDENTS

40,000+

Fatal road traffic incidents recorded annually across US highway networks.

PEAK HAZARD WINDOW

3–5 PM

Evening commute surge generating peak daily accident frequencies (~12,500 cases).

ECONOMIC COST

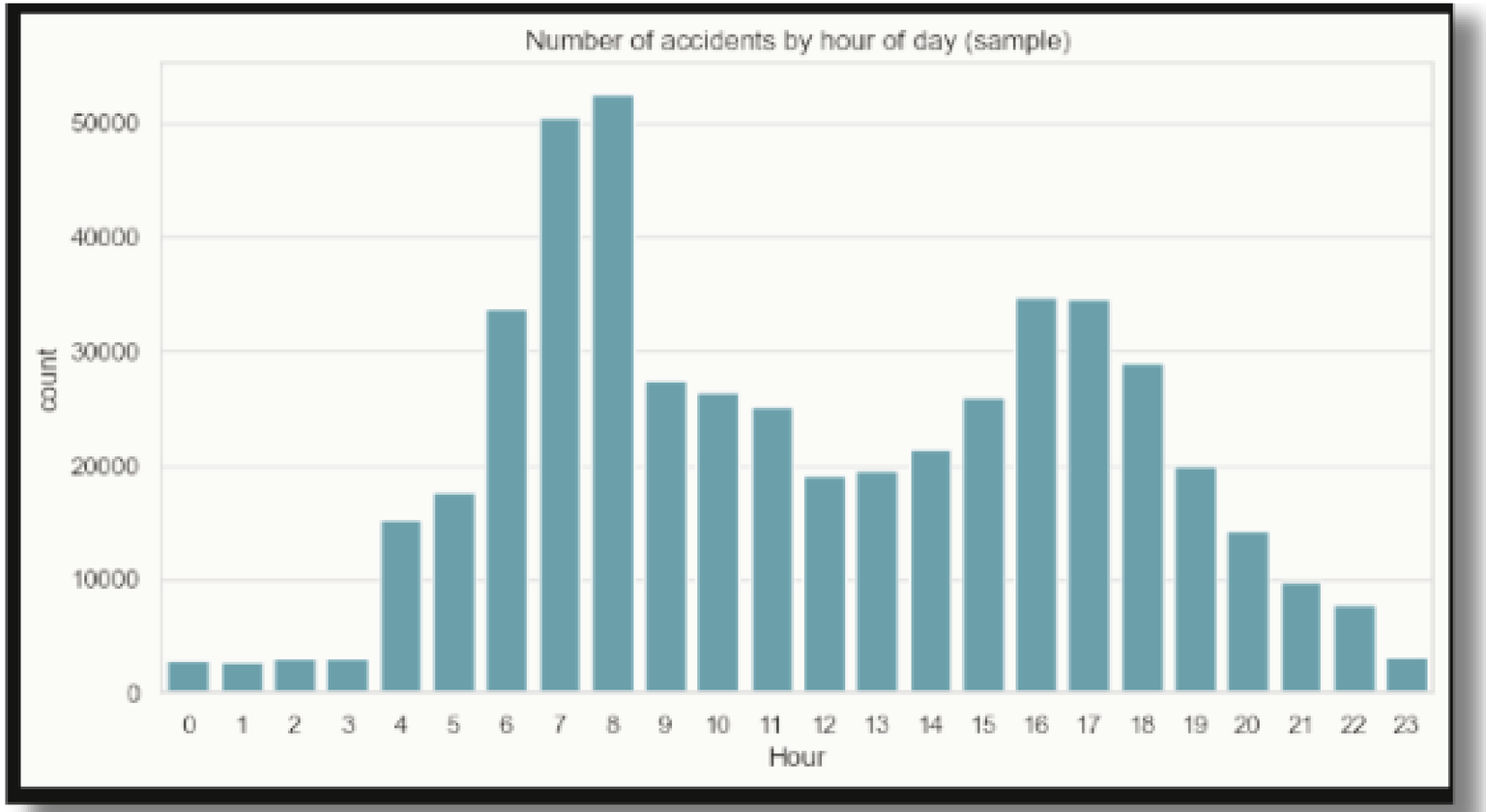
$430B

Estimated annual economic burden from medical expenses, property loss, and delay.

When Do Accidents Happen Most?

Friday High: 17.2%

Fridays record the highest proportion of weekly accidents, followed by Thursday (16.1%).



Low Risk Morning: 4–6 AM

Early morning hours show the lowest crash frequency across all road functional types.

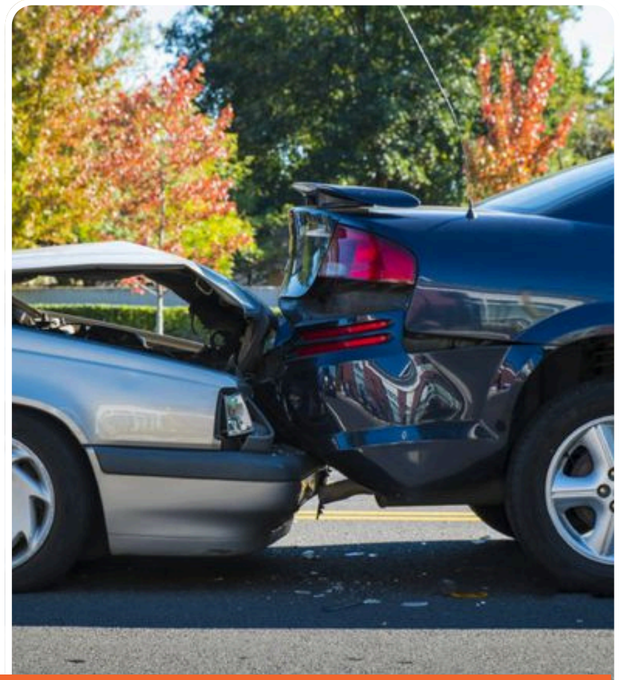
Winter Season Spikes

Cold winter months experience heightened accident rates due to weather and low daylight.

Major Drivers of Traffic Accidents

Rear-End Collisions

"Same Direction Rear End" is the single most common collision type, accounting for ~50,000 recorded incidents.



Lane Switching & Turns

"Switch Movement Angle" crashes closely follow as the second highest category (>30,000 incidents).

Weather Dynamics

Rain and visibility constraints amplify severity, though driver caution lowers total counts during heavy snow.

Data-Driven Recommendations for Drivers

↔️ Increase Following Distance

Mitigates the #1 hazard: same-direction rear-end collisions during peak traffic.

🕑Heighten Vigilance 3 PM – 5 PM

Stay focused during high-risk rush hour periods when accident rates peak.

🗓️Plan Weekend Commutes Early

Exercise extra caution on Thursday and Friday afternoons due to elevated traffic density.

❄️Adapt Driving to Low-Light & Winter Conditions

Turn on headlights early during dusk, decrease speed by 10–15 mph on wet or icy roads, and inspect tire tread before the winter season starts.

BUILDING SAFER ROADS THROUGH DATA INSIGHTS

Thank You

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